

Project Overview

The Science Impact Collaborative CBA Dashboard

Our goal is to develop and launch a comprehensive, publicly accessible dashboard focused on Community Benefits Agreements (CBAs) in the context of renewable energy and infrastructure development. This tool will serve as a practical resource for municipal leaders, community organizers, and local government officials navigating the complex landscape of community-engaged development processes.

Our vision is to create a go-to digital resource that empowers communities and municipalities to design, negotiate, and implement effective and enforceable CBAs for renewable energy and infrastructure projects. The dashboard should be:

Practical: Providing concrete templates, checklists, and process guidance

Accessible: User-friendly interface requiring no technical expertise

Contextual: Allowing users to navigate based on their role, project type, and local circumstances

Evidence-based: Grounded in research while incorporating practitioner wisdom

Living: Designed for iteration and improvement based on user feedback

Intended Impact

This dashboard will serve distinct stakeholder groups with different but complementary needs. Municipal leaders - town managers, sustainability coordinators, and planning directors - will gain efficient access to reliable guidance and templates, allowing them to build confidence in navigating CBA processes while learning from peer communities' experiences. Community organizers and environmental justice advocates, often working in volunteer or part-time capacities, will find tools to level the playing field in negotiations with developers, understanding what benefits are realistic based on what others have achieved and accessing frameworks to structure and amplify community input. Developers will benefit from understanding community expectations earlier in project timelines, reducing conflict and delays through proactive engagement while accessing models of successful CBAs that benefited all parties.

The systemic impact extends beyond individual projects. By equipping communities to better advocate for their needs, we enable more equitable development outcomes. By reducing project delays stemming from community opposition, we help accelerate the clean energy transition at a moment when speed matters. The dashboard will strengthen social license for renewable development by improving trust between communities and developers, while rapidly diffusing best practices across municipalities that might otherwise operate in isolation.

Concrete Goals

We will develop a web-based, interactive dashboard hosted on the **Science Impact Collaborative website** with clear navigation based on user type (municipal official, organizer,

developer, researcher) and project stage (early engagement, negotiation, implementation, monitoring). The dashboard will include core content modules covering CBA fundamentals, benefit categories with examples, process frameworks for stakeholder identification and negotiation, case studies with lessons learned, and practical templates including sample agreements and meeting facilitation guides. Interactive features will include a filterable database of CBA provisions searchable by project type and benefit category, self-assessment tools, a resource library linking to complementary tools and organizations, and comparison functionality allowing users to examine benefit structures across similar projects. The dashboard will be mobile-responsive, WCAG-compliant for accessibility, and integrated with Science Impact Collaborative branding.

Success will be measured across three time horizons. In the first six months, we aim for 500+ unique visitors with average session durations exceeding three minutes, 50+ template downloads, and positive feedback from municipal beta testers. Medium-term success (6-18 months) means reaching 2,000+ unique visitors, documented use in at least three real CBA processes, invitations to present the tool at conferences or municipal associations, and requests for customization from other regions. Long-term (18+ months), the dashboard should be established as a recognized field resource—cited in reports, linked by peer organizations, continuously iterated based on user feedback, and potentially adapted to infrastructure contexts beyond renewable energy.

Target Audiences

Our primary audiences are municipal officials and staff working in small to mid-sized municipalities (population 5,000-100,000) who are often handling renewable energy siting for the first time with limited staff capacity for extensive research. Community organizers and leaders, including environmental justice advocates and neighborhood association representatives, typically work in volunteer or part-time capacities and may lack legal or technical expertise in development processes, though they possess strong knowledge of community needs and priorities. Local government boards, including selectboards, planning boards, and energy committees, occupy volunteer positions with limited time yet bear responsibility for decisions affecting their communities, needing clear frameworks to balance multiple stakeholder interests.

Secondary audiences include developers and project proponents seeking to build community support and reduce project risk through better understanding of community expectations and best practices. Consultants and technical advisors supporting municipalities or communities in CBA processes need comprehensive reference materials and case studies. Finally, researchers and students studying community engagement, energy transitions, or planning processes require curated information and examples for academic work. While we design primarily for municipal and community users, serving these secondary audiences extends the dashboard's impact and helps ensure CBA knowledge reaches those who influence project outcomes.

Proposed Workplan

Project Approach

February 2025

Week 1-2 (Feb 10-23):

- Complete landscape analysis framework and begin tool reviews (Lucy) (DONE)
- Create interview contact spreadsheet with initial contacts (Lucy) (DONE)
- Update project overview document with sources and references (Sanjana) (DONE)
- Design interview guides for different stakeholder categories (Sanjana) (DONE)
- Draft and submit IRB exemption application (Jung, with Sanjana's abstract) (DONE)

Week 3-4 (Feb 24 - Mar 9):

- Complete comprehensive landscape analysis (DONE)
- Analyze landscape analysis findings and identify key gaps (Lucy) (DONE)
- Draft initial dashboard structure based on research (Lucy) (DONE)
 - Review findings from the paper and apply as needed.
 - Add core interview questions to the interview guide based on the dashboard ideation.
- Review interview guides for different stakeholder categories (Team) (DONE)
- Refine interview guides based on landscape analysis findings (Sanjana) (DONE)

March 2025

Weeks 1-2 (Mar 10-23):

- Finalize interview contact list across all stakeholder categories (Team) (DONE)
- Begin outreach to interview participants (Lucy, with team support) (DONE)
- Begin conducting stakeholder interviews (target 3-4 per category) (Lucy lead) (DONE)

Weeks 3-4 (Mar 24 - Apr 6):

- Complete remaining stakeholder interviews (Lucy)
- Synthesize interview findings with landscape analysis (Lucy)
- Define dashboard features, content areas, and user flows (Team)

April 2025

- Begin content development for core dashboard sections (Lucy)
- Integrate policy implications from published CBA paper (reference Jung's suggestions)
- Create text-based versions of: state legislation guidance, capacity building resources, monitoring guidelines
- Develop resource aggregation section (links to other tools, databases, facilitation support)

May 2025

- Complete all dashboard content sections in text format (Lucy)
- Create flow charts and information architecture diagrams (Lucy)

- Develop content for: municipal guidance, community toolkit sections, developer resources
- Finalize Phase 1 deliverables and prepare handoff materials for summer digitization

Phase 1 Deliverables (End of Spring 2025)

Comprehensive landscape analysis document

- Analysis of existing CBA tools and resources
- Gap analysis identifying opportunities
- Audience and feature comparison

Stakeholder research summary

- Interview findings across all stakeholder categories
- User needs and preferences analysis
- Validated assumptions and revised approach

Complete dashboard content (text-based)

- All core content sections written and organized
- Municipal guidance materials
- Community toolkit resources
- State legislation and policy guidance
- Capacity building and monitoring resources
- Aggregated links to external resources

Information architecture and user flow diagrams

- Site structure and navigation design
- User journey maps for different audiences
- Content organization and hierarchy

Technical specifications for digitization

- Feature requirements and specifications
- Content formatting guidelines
- Handoff documentation for undergraduate developers

Phase 2: Digitization (Summer 2025)

Timeline: June - August 2025

- Recruit and train undergraduate developers
- Transform text content into web-based dashboard
- Implement user interface and navigation
- Create visual elements, diagrams, and interactive features
- Integrate with SIC clinic website architecture

Phase 3: Testing & Launch (Late Summer/Fall 2025)

Timeline: August - September 2025

- Beta testing with original interview participants
- Gather feedback and implement final revisions
- Prepare launch materials and user documentation

- Public launch by end of September 2025
- Lucy may return for final refinements and launch support

Roles and Responsibilities

Lucy (Spring 2025):

- Lead researcher for landscape analysis and stakeholder interviews
- Primary content developer for all dashboard sections
- Create information architecture and user flow diagrams
- Prepare handoff documentation for summer development

Sanjana:

- Project planning and timeline oversight (February focus)
- Interview guide design and stakeholder identification
- Monthly check-ins and strategic guidance throughout
- Integration of existing research and policy implications

Jung:

- Project coordination and timeline management
- IRB application and administrative oversight
- Summer undergraduate RA recruitment and management

Abstract!!

Community Benefits Agreements (CBAs) in renewable energy development lack accessible, practical resources to help municipalities and communities navigate CBA processes effectively. This study will inform the development of a web-based dashboard serving municipal leaders, community organizers, and local government officials.

The research employs anonymous virtual interviews with stakeholders across four categories: municipal officials, community organizers, developers, and existing CBA tool creators. Interviews will explore current challenges in CBA processes, user needs for guidance and templates, gaps in existing resources, and preferences for digital tool design. A landscape analysis will systematically review existing CBA resources to identify content gaps and opportunities for innovation.

All interviews will be conducted virtually and anonymously, with no personally identifiable information collected. Participants will not be quoted directly, and findings will be reported only in aggregate. The research aims to create evidence-based tools that strengthen community voice in renewable energy development while reducing project conflicts.